US–Iran War Data Analysis

2026 • Project Documentation

Project Focus
Analysis of reported conflict events, military attacks, casualties, geographical patterns, humanitarian impacts, and economic indicators associated with the US–Iran conflict.

A data analytics project using Python, Pandas, Matplotlib, and Power BI to turn source data into structured analysis, visual evidence, and interactive dashboards.

Data sources proposed: IranWar.ai research dataset, supplemented where available by ACLED and United Nations reports.

Documentation note: This document describes the proposed analysis workflow and dashboard design. It does not claim that specific casualty totals, event counts, or economic findings have already been verified.

1. Project Overview

This project analyzes data related to the US–Iran conflict in 2026, with attention to recorded military events, attacks, reported casualties, affected locations, humanitarian consequences, and selected economic indicators.

The main goal is:

Transform raw conflict-related data into meaningful, traceable insights through data cleaning, exploratory data analysis (EDA), and interactive dashboards.

The analysis is intended to help users explore how reported events vary over time and place, compare reported deaths and injuries, examine event categories, and view potential temporal relationships between conflict activity and economic indicators.

Tools: Python, Pandas, Matplotlib, and Power BI.

Proposed sources: IranWar.ai research dataset; ACLED and UN reports where suitable data are available and can be compared responsibly.

2. Main Analysis Pipeline

```
Source datasets
↓
Inspect files, schemas, and source metadata
↓
Data profiling and quality checks
↓
Cleaning, standardization, and validation
↓
Combine compatible event, casualty, geography, and economic data
↓
Exploratory Data Analysis (EDA)
↓
Calculate metrics and document limitations
↓
Build charts and Power BI data model
↓
Interactive dashboard + evidence-based findings
```

3. Data Management & Preparation

The exact steps depend on the structure, coverage, and licensing of the downloaded files. Each source should be inspected before it is combined with another.

Potential input data

- Conflict/event records: event date, location, country, event type, actors or parties when available, and source reference.
- Casualty reports: fatalities, injuries, civilian/military classification, reporting date, and source, where available.
- Geographical data: country, region, administrative area, latitude, and longitude when provided or verifiable.
- Humanitarian indicators: reported infrastructure damage and affected populations, only where supported by a source.
- Economic time series: oil prices and selected financial-market indicators with dates and units.

Data profiling

- Inspect column names, data types, row counts, date ranges, and geographic coverage.
- Measure missingness, duplicate records, invalid dates, inconsistent category labels, and unusual values.
- Check whether event records and casualty records represent unique events, reports, or updates.

Data cleaning and validation

- Standardize date formats, country/region labels, event categories, and numeric fields.

- Preserve original source fields and create documented cleaned fields rather than silently overwriting source values.
- Flag duplicate or potentially overlapping reports for review; do not automatically assume similar records are the same event.
- Validate latitude/longitude ranges and ensure dates and units align across sources.
- Keep a cleaning log recording each change, rule, and unresolved issue.

Important limitation

Reported figures may differ between sources because of timing, definitions, access restrictions, and verification standards. Retain source attribution and distinguish confirmed, reported, estimated, and unknown values whenever the data allow it.

4. Analytical Data Model

A simple model can keep event-level analysis separate from casualty, humanitarian, geography, and economic data.

```
Fact Conflict Events
• Event ID / source record ID
• Event date and reporting date (if available)
• Event type and sub-type
• Location / country / region
• Reported fatalities and injuries (if event-linked)
• Source and verification/status fields

Related dimensions / tables
• Date
• Geography
• Event type
• Source / report
• Casualty reports (if separate)
• Humanitarian impact records (if separate)
• Economic indicators (date-indexed)
```

Use a star-schema or a small set of clearly related tables in Power BI when the source data support reliable keys. Avoid joining records only because they share a country or date; that can multiply counts or imply relationships the data do not establish.

5. Core KPIs & Summary Dashboard

The dashboard should calculate metrics from the available, cleaned data. Each KPI should display its definition and reporting coverage.

KPI	Definition / interpretation
Recorded events	Count of unique event records under the documented deduplication rule.
Reported fatalities	Sum of valid reported fatality values; show source and date coverage.
Reported injuries	Sum of valid reported injury values where available.
Affected countries	Distinct count of countries with at least one included event.
Events over time	Event count by day, week, or month depending on data volume.
Civilian infrastructure impacts	Count or sum of source-defined damage indicators, only if recorded consistently.
Economic indicators	Date-aligned oil-price and selected market measures, shown separately from conflict-event totals.

Do not combine fatalities from separate sources without checking overlap. If unique event IDs are unavailable, clearly label the totals as source-record counts or reported totals rather than definitive counts of unique incidents.

6. Conflict Timeline Analysis

Purpose: understand how the number and type of reported events change over the available reporting period.

Recommended visuals

- Line chart: recorded events per day/week/month.
- Stacked column chart: event counts by type over time, if categories are consistent.
- Optional cumulative trend: cumulative recorded events, with a clear note that it reflects the dataset's records and coverage.

Questions to explore

- When do recorded events peak or decline?
- Do changes differ by event type or affected country?
- Are apparent changes potentially explained by changes in source coverage or reporting frequency?

Use a consistent time grain and show gaps in the reporting period rather than silently treating missing dates as zero events.

7. Casualty Analysis

Purpose: compare reported fatalities and injuries while preserving differences in source definitions and confidence.

Recommended visuals

- Bar chart: reported fatalities by country or region.
- Separate bar chart: reported injuries by country or region.
- Stacked chart by civilian/military category only if the source includes comparable classifications.
- Timeline of reported casualties, using the same time grain and definitions across compared groups.

Interpretation rules

- Label all values as reported casualties unless the source establishes another status.
- Show unknown or unclassified categories separately where useful.
- Do not infer civilian or military status when it is absent.
- Avoid ranking countries when coverage, reporting period, or definitions are not comparable.

8. Geographical Analysis

Purpose: show where events are recorded and how their distribution varies across affected areas.

Recommended visuals

- Map of event locations using valid coordinates or appropriately geocoded administrative areas.
- Country/region choropleth when a meaningful comparable measure exists.
- Ranked bar chart of event counts by country or region as an accessible alternative to a map.

Data checks

- Verify coordinate order and coordinate validity.
- Do not geocode vague place names as if they were precise locations.
- Use aggregation or clustering where many points overlap.
- Include a clear note when location is approximate or only available at country/region level.

9. Event Type & Military Incident Analysis

Purpose: describe the composition of the events represented in the dataset.

- Bar chart of records by event type and sub-type.
- Cross-tabulation of event type by country or region.
- Trend chart for selected event categories.
- Optional event-type versus reported-casualty comparison only where the linkage and definitions are valid.

Use source-defined labels where possible. If categories from different datasets are not equivalent, preserve each source's taxonomy or document the mapping.

10. Humanitarian Impact

Purpose: summarize documented effects on civilians, infrastructure, and affected populations where data are available.

- Infrastructure damage by source-defined category, location, and time.
- Reported affected-population measures, retaining the original unit and definition.
- Separate civilian-impact indicators from military event counts.
- Display coverage gaps and the date of the latest available update.

This page should not imply that the dataset captures all humanitarian consequences. Missing observations must not be interpreted as no damage or no affected people.

11. Economic Impact & Market Indicators

Purpose: explore how selected economic indicators move during the period under study, without claiming that conflict events alone caused those changes.

- Line chart of oil prices over time.
- Line charts or indexed comparisons for selected financial-market indicators, with clear units and source notes.
- Optional event timeline annotations to compare dates of reported events with market movements.
- Date slicer to select the conflict period and a comparison period where data support it.

Analytical caution

Temporal co-movement is not proof of causation. Market prices respond to multiple geopolitical, supply, demand, policy, and financial factors. Describe results as observed relationships or timing comparisons unless a defensible causal design is performed.

12. Exploratory Data Analysis (EDA)

Recommended workflow

- Univariate analysis: event counts, distributions, missing values, and category frequencies.
- Bivariate analysis: event type by country, event counts by time, and casualties by category where comparable.
- Time-series analysis: trends, peaks, gaps, and changes in reporting volume.
- Geospatial analysis: event concentration and regional differences at the available level of precision.
- Data-quality analysis: source coverage, duplicate candidates, late updates, and conflicting reported values.

Every finding should be traceable to a defined metric, a date range, a filter state, and the source records used.

13. Power BI Dashboard Design

Suggested dashboard pages

Page	Main content
1. Executive Overview	KPI cards: recorded events, reported fatalities, reported injuries, affected countries; event trend; top event types; coverage note.
2. Conflict Timeline	Events over time, event type trends, peak periods, date filters.
3. Casualty Analysis	Separate fatalities and injuries views; country/region comparisons; categories if supported.
4. Geographic Distribution	Map or regional chart, event density, country/region filters, location-precision notes.
5. Event Types	Event-type breakdown, trends, and country/region comparison.
6. Humanitarian Impact	Infrastructure damage and affected populations where data exists; source coverage and limitations.
7. Economic Indicators	Oil prices and selected financial indicators over time, with event-date context and non-causal interpretation.

Recommended slicers

- Date range.
- Country / region.
- Event type / sub-type.
- Source.
- Casualty category or verification status, if available.

Keep slicers consistent across pages where the fields share the same meaning. Add a visible data coverage/source note and avoid displaying a KPI when its underlying field is unavailable or not comparable.

14. Data Sources, Attribution & Evidence

Proposed source strategy

- IranWar.ai research dataset: use the supplied schema, metadata, definitions, and collection notes; verify its coverage before analysis.
- ACLED: use only relevant records and fields that are accessible under the applicable terms and are suitable for the question.
- United Nations reports: use for corroboration or contextual indicators where definitions, dates, and geographic coverage align.

Source documentation checklist

- Record the source name, dataset/report title, access or publication date, URL, and license/terms.
- Record the reporting period, geographic coverage, event definitions, and known limitations.
- Retain source references for each record or group of records when possible.
- Distinguish source-derived values from calculated metrics and analyst interpretations.

No exact dataset totals are included in this documentation because the underlying conflict dataset was not supplied with the project brief.

15. Data Quality, Ethics & Interpretation

- Use neutral, source-attributed language and avoid presenting contested reports as independently verified facts.
- Separate event date from publication/report date when both exist.
- Document revisions and avoid double-counting reports that describe the same event.

- Keep uncertainty, unknown categories, and missing data visible.
- Avoid exposing sensitive personal details or unnecessary exact locations of vulnerable people.
- Do not infer responsibility, intent, or causation beyond what the data and source documentation support.
- Provide a short methodology and limitations note on each dashboard page.

16. Main Dashboard Interactions

Users should be able to

- Filter by date range, country/region, event type, and source where available.
- Cross-filter charts by selecting a country, category, or time period.
- Compare reported fatalities and injuries without merging their definitions.
- Move from overview metrics to detailed trends and source context.
- Reset filters and inspect the current reporting coverage.
- Export selected visuals or report pages where Power BI permissions allow.

17. Recommended Technology Stack

Component	Technology / role
Data collection	Source downloads / documented data extracts
Data processing	Python, Pandas, NumPy
EDA & static charts	Matplotlib
Data validation	Pandas checks and documented validation rules
Analytical storage	CSV/Parquet or a relational database if scale requires it
Interactive dashboards	Microsoft Power BI
Geospatial visualization	Power BI map visuals, subject to data quality and configuration
Version control	Git + GitHub
Documentation	README, data dictionary, source log, methodology notes

The stack can remain lightweight: Python is used for cleaning and EDA, while Power BI presents the interactive dashboard. A database or additional tools should be added only if project scale or requirements justify them.

18. Project Workflow & Deliverables

Phase	Tasks	Deliverable
1. Source review	Inspect source files, schemas, coverage, and terms.	Source inventory and data dictionary.
2. Cleaning	Profile, standardize, validate, and document issues.	Cleaned dataset and cleaning log.
3. EDA	Analyze trends, distributions, categories, casualties, and coverage.	Notebook/script and charts.
4. Metric design	Define KPIs, counting rules, and comparison limits.	Metric definitions.
5. Power BI model	Build relationships and measures; test filter behavior.	Power BI data model.
6. Dashboard	Create pages, slicers, and source/limitation notes.	Interactive dashboard.
7. Validation	Reconcile counts against source records and test edge cases.	QA checklist and findings.
8. Final documentation	Describe methods, results, caveats, and reproducibility.	PDF report and project README.

19. Example Questions the Analysis Can Answer

- How many event records are present in the selected reporting period?
- How have recorded events changed over time?
- Which countries or regions have the largest number of recorded events in this dataset?
- How do reported fatalities and injuries vary by country, event type, or period?
- Which event types are most frequently recorded?
- What humanitarian impact indicators are available, and where are the coverage gaps?
- How did oil prices and selected financial indicators move during the same period?
- Which patterns are robust across sources, and which depend on reporting coverage or definitions?

20. Expected Project Value

The project brings together four complementary areas:

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Data Engineering
Raw source files → documented, validated analytical data

Exploratory Data Analysis
Analytical data → trends, comparisons, and quality checks

Data Visualization
Metrics and patterns → understandable charts and maps

Business Intelligence
Validated measures → interactive Power BI exploration
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Together, these components provide a structured way to explore reported conflict events and their associated human and economic impacts, while making source limitations and uncertainty visible.

21. Final Notes & Limitations

- The project brief identifies intended sources but does not include the underlying datasets, so exact totals and findings are not asserted here.

- Source availability, access permissions, licensing, field definitions, and update frequency must be checked before analysis.
- Economic comparisons should be described as associations in time, not causal conclusions.
- Casualty and humanitarian figures should retain their source attribution and uncertainty.
- Final dashboard metrics must be adjusted to the actual columns, quality, and coverage of the selected data.